

1.6d

# Sensation

Skin, Chemical & body Senses & sensory interaction

## What is touch?

affectionate touches are essential while developing into an adult

"sense of touch" is a mixture of:

- pressure
- warmth
- cold
- pain

Some spots on our skin are especially sensitive to some of these.

Other sensations are a combination of these

ex.

stroking adjacent pressure spots → tickle  
touching adjacent cold & pressure spots → wetness  
repeated gentle stroking of pain spots → itchiness  
activating receptors for cool & warm → hotness

touch sensations involve more than tactile stimulation

ex. self-tickling makes less somatosensory cortex activation than the same tickle from someone else.

a leg caress evokes a different somatosensory cortex response when a straight man believes it comes from a woman rather than a man.

## How is pain affected?

Pain is important to alert you that something is wrong.

Pain also enhances self-awareness, arouses empathy & promotes social connections

Those born unable to feel pain are at higher risk of severe injury or early death

Pain reflects both bottom-up sensations & top-down cognitions.

Pain is a biopsychosocial phenomenon.

---

## Biological influences

Pain is a physical event produced by senses

But pain differs from other senses

- no one kind of stimulus trigger pain like light triggers vision
- no specialized receptors process pain signals the way retina receptors react to light rays

nociceptors detect harmful temperatures, pressure or chemicals

② mostly in skin but also in muscles & organs

Pain depends on inherited genes & physical characteristics, partly

Women are more sensitive to pain

Gate-Control theory:

Spinal cord contains neurological 'gate' controlling transmission of pain to brain

Small spinal cord nerve fibers conduct most pain signals

Injury activates the small fibers & opens gate

pain signals then travel to brain → you feel pain

large-fiber activity, simulated by massage, electrical stimulation, or acupuncture. Can close the pain gate by blocking pain signals

∴ Chronic Pain can be treated by gate-closing stimulation, ex. massage, and mental activity, ex. distraction

Endorphins: Our own natural painkillers

① released in response to severe pain or vigorous exercise.

ppl w/ genes boosting endorphins have less pain

ppl w/ gene mutation disrupting pain circuit transmission may not feel pain

## Phantom Pain

Brain can create pain, like in phantom limb sensations

w/o normal sensory input from missing limb, brain misinterprets & amplifies spontaneous but irrelevant CNS activity,

2/10 limb amputees feel pain or movement in nonexistent limbs

Even those born w/o a limb can sometimes perceive sensations from that limb

deaf ppl may experience Tinnitus  $\therefore$  phantom sound of ringing in ears

People who lost vision to some conditions may experience phantom sights - harthrreathening hallucinations.

Others w/ nerve damage in tasting & smelling systems have experienced phantom tastes or smells, ex. ice water that's very sweet or fresh air reeking of rot

## Psychological influence on pain

Attention we focus on it is a powerful influence on pain

We edit our memories of pain

people overlook pain's duration, focusing on peak pain & pain at the end

ppl. who immersed one hand in painfully cold water for 60 sec, and other in same water for 60 sec + 30 more sec of less hot water preferred the longer trial.

we prefer longer pain if there is less pain at the end.

end of experience can affect pleasure memory too

ppl told it was their last piece of chocolate liked it more than ppl told it was their next

## Sociocultural influence on pain

We perceive more pain when others also seem to be experiencing pain.

When ppl feel empathy for someone's pain, brain activity partly mirrors brain activity of the actual brain in pain.

## Controlling Pain

∴ pain is physical and psychological phenomenon  
∴ it can be treated physically & psychology

Pain control therapies may incl.

drugs, surgery, acupuncture, electrical stimulation, massage, exercise, hypnosis, relaxation training, meditation & relaxation training

Placebos can mimic painkilling drugs & even cause side effect.

Distraction :: drawing attention away from painful stimulus, is an effective way to inhibit pain & increase pain tolerance.

## Taste

Gustation involves 6 basic sensations

Umami is a savory meaty taste;  
monosodium glutamate

sweet

sour

salty

bitter

Umami

Oleogustus

Oleogustus, the unique taste of fat

Pleasureful tastes attracted ancestors to energy- or protein-rich foods, enabling survival

Aversive tastes deterred them from new foods that may be toxic

∴ 2-6 year olds are fussy eaters, especially for new foods.

Given small, repeated tastes of disliked but safe new foods, most begin to accept.

Babies fed vanilla milk grow up to prefer vanilla flavor

| Taste | Indicates |
|-------|-----------|
|-------|-----------|

Sweet

Energy Source

Salty

Sodium essential to physiological process

Sour

Potentially toxic acid

Bitter

Potential Poisons

Umami

Proteins to grow, repair tissue

Oleogustus

Fats for energy, insulation, cell growth.

Taste is chemical

In each bump on the top & side of tongue  
≥ 200 taste buds

① contains a pore that catches food chemicals & releases neurotransmitters



Each pore has 50-100 taste receptor cells

① project antenna-like hairs that sense food molecules

receptors are especially sensitive to one kind of taste

Each receptor transmits to matching partner cell in temporal lobe.

Some ppl have more taste buds, enabling them to experience more intense tastes.

They are superfasters, compared to the average-ability medium tasters & the lower-than-average nontasters

taste receptors reproduce ~ every 1-2 weeks

age  $\uparrow$   $\rightarrow$  # taste buds  $\downarrow$ , taste sensitivity  $\downarrow$

Taste is psychologically influenced by expectations.

# Smell

On average we breathe 500 mil breaths

Olfaction :: our sense of smell

Smell is chemical

Smelling :: molecules of substance carried in air reach tiny cluster of receptor cells at the top of each nasal cavity

Sniffing :: Swirl air up to those receptors, enhancing aroma

Olfactory receptors respond selectively to specific sets of odor, then alert brain quickly through their axon fibers  
① 20 million

Olfaction is a very old, primitive sense that predates cerebral cortex  
∴ Olfactory neurons bypass the thalamus.

Pheromones :: olfactory chemical messengers.

Some pheromones serve as sexual attractions.

~350 receptor proteins that recognize particular odor molecules.

receptor proteins are embedded on the surface of nasal cavity neurons

Odor molecules slip into these like key into lock

We don't have distinct receptor for each detectable odor.

Odors trigger combinations of receptors, in patterns interpreted by olfactory cortex.

Odor molecules bind to different receptor arrays producing up to 1 trillion odors we can discriminate

Smelling romantic partner causes stress to drop

Our sense of smell is weak

Smells are harder to describe & recall

We are good at recognizing long forgotten odors & their associated memories.

Hotline between brain area receiving info from nose & the limbic centers associated with memory & emotion

How do we detect our position?

Moving forward requires feedback from & instructions to ~200 muscles & it engages brain power exceeding brain activity from reasoning

millions of proprioceptors :: position & motion sensors in muscles, tendons and joints :: provide constant feedback to brain

This enables sense of kinaesthesia, keeping you aware of body position & movement

Vision interacts with kinaesthesia; difficult to balance without sight.

Vestibular sense monitors head's (i.e. body's) position & movement

2 biological gyroscopes for this in inner ear

Semicircular canals, fluid fills, looks like 3d pretzel

Vestibular sacs, pair, calcium-crystal filled

head rotates, tilts → movement of these organs stimulates hair-like receptors, send nerve signals to cerebellum, enabling you to sense your position & maintain balance

a twirl & abruptly halt:  
kinesthetic receptors & fluid in semicircular canals will not return to normal.

This creates dizziness.

Vestibular sense is very fast.

Slip ⇒ vestibular sensors automatically order skeletal response before you consciously decide to right yourself.

| Sensory System                         | Source                                                           | Receptors                                                                                         | Key Brain Areas              |
|----------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|
| Vision                                 | light waves strike eye                                           | Rods & cones in Retina                                                                            | Occipital lobe               |
| Hearing                                | Sound waves strike outer ear                                     | Cochlear hair cells (cilia) in inner ear                                                          | Temporal lobe                |
| Touch                                  | Pressure, warmth, cold, harmful chemicals                        | Receptors (incl. pain-sensitive nociceptors), mostly in skin, detect pressure, warmth, cold, pain | Somatosensory Cortex         |
| Taste                                  | Chemical molecules in the mouth                                  | Basic taste receptors for sweet, sour, salty, bitter, umami, oleogustus                           | Frontal/Temporal lobe border |
| Smell                                  | Chemical molecules breathed in through the nose                  | Millions of receptors at the top of the nasal cavities                                            | Olfactory bulb               |
| Kinesthesia<br>position & movement     | Any change in position of a body part, interacting with vision   | Kinesthetic sensors in the joints, tendons, and muscles (proprioceptors)                          | Cerebellum                   |
| Vestibular Sense<br>balance & movement | Movement of fluids in the inner ear caused by head/body movement | Hair-like receptors (cilia) in the ears' semicircular canals & vestibular sacs                    | Cerebellum                   |

# Sensory Interaction

All senses 'eaves drop' on another :: Sensory interaction

A major part of taste is smell

Texture can also influence taste

Visuals & sounds affect each other

If we see a speaker form one word with their word, but hear a different, our brain may combine both into a third new word.

Brain circuits processing physical sensations interact with circuits for cognition.

This results in embodied cognition

ex. Sitting at wobbly desk may make relationships seem less stable  
People who sit on harder chairs give harsher punishments

For individuals w/ synesthesia, stimulation of one sense triggers experience of another.

ex. Seeing numbers may evoke taste or color sensations

MCA

1a 2d 3a 4a 5b 6